

Investigating the Effects of Artificial Intelligence-Assisted Language Learning Strategies on Cognitive Load and Learning Outcomes: A Comparative Study

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Abstract

This study investigates the impact of AI-assisted language learning (AIAL) strategies on cognitive load and learning outcomes in the context of language acquisition. Specifically, the study explores three distinct AIAL strategies: personalized feedback and adaptive learning, interactive exercises with speech recognition, and intelligent tutoring with data-driven insights. The research employs a pretest-posttest random assignment experimental design, utilizing three experimental groups and a control group, with a total of 484 EFL students specializing in teaching English as a foreign language participating in the study. Data collection involves pre- and post-tests, questionnaires, and interviews to assess the influence of AIAL strategies on cognitive load and learning outcomes. Cognitive load is measured using the Cognitive Load Scale, while pretest-posttest assessments evaluate the efficacy of AIAL interventions across various language skills. These results contribute to the existing body of AIAL research by offering empirical evidence for the effectiveness of specific strategies in optimizing language learning experiences. The implications of this study extend to educators, researchers, and developers in the field of AIAL, emphasizing the potential of AIAL to enhance language acquisition processes and inform instructional design practices.

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Keywords

Artificial intelligence-assisted language learning, cognitive load, personalized feedback, adaptive learning, speech recognition, intelligent tutoring

Introduction

The integration of artificial intelligence (AI) in the field of language learning has attracted significant attention in recent years due to its potential to enhance learning outcomes and facilitate personalized learning experiences (Yanhua, 2020). Researchers have extensively explored the impact of AI-assisted language learning (AIAL) from various perspectives. For example, Lin et al. (2021) investigated the structural relationship between primary students' motivation to learn AI, identifying intrinsic motivation, perceived competence, and perceived usefulness as significant predictors of motivation in AI learning. These findings suggest that students' willingness to engage in AI learning may be influenced by their perception of competence and usefulness, carrying important implications for the design and development of effective AIAL strategies.

Additionally, Weng and Chiu (2023) conducted a systematic review in the field and discovered that instructional design factors such as learner-centeredness, interactivity, and adaptivity positively impacted learning outcomes in AI-assisted language learning. The study by Huang et al. (2023) conducted a comprehensive review of the trends, research issues, and applications of AI in language education. The review uncovered the extensive use of AI in language assessment, teaching, and learning support. However, the review also identified research gaps, including the need for more empirical studies to assess the effectiveness of AI-based language education systems. Furthermore, ethical and privacy concerns were highlighted as crucial considerations in the design and development of these systems.

In a related study, Huang et al. (2022) systematically reviewed the effectiveness of chatbot-supported language learning. Their analysis aimed to evaluate the usefulness of chatbots in language learning contexts and identify their benefits and limitations. The review revealed the potential of chatbots as a valuable tool for providing personalized and interactive language practice. However, the study also highlighted certain limitations and challenges, such as accuracy, contextual understanding, and learner motivation.

The study by Zhai and Wibowo (2023) conducted a systematic review on the use of artificial intelligence dialogue systems to enhance English as a foreign language students' interactional competence in university settings. The research examined the effectiveness of these systems in promoting meaningful and authentic conversations in English. Through a comprehensive analysis of relevant literature, the review highlighted the potential of artificial intelligence dialogue systems to enhance English as a Foreign Language (EFL) students' language learning experiences by providing opportunities for authentic and interactive language practice.

In another study, [Liu et al. \(2021\)](#) investigated the integration of a reflective thinking promoting mechanism into artificial intelligence-supported English writing environments. Their study aimed to examine the effectiveness of incorporating this mechanism in enhancing English writing skills. Through an analysis of the interactive learning environments, the authors explored the impact of the reflective thinking promoting mechanism on students' writing performance.

Additionally, [Huang et al. \(2023\)](#) explored the role of AI in providing personalized feedback and supporting adaptive learning in language education. Through an examination of existing literature, the authors identified the potential of AI technologies to deliver personalized feedback to language learners, catering to their individual needs and promoting targeted language improvement. The study highlighted the use of AI in adaptive learning environments, where AI algorithms can analyze learner data and dynamically adjust the learning content and activities to meet the unique learning needs of each individual.

[An et al. \(2023\)](#) conducted a study on modeling students' perceptions of artificial intelligence assisted language learning. The study aimed to explore how students perceive the use of AI in language learning and its impact on their learning experiences. Through a comprehensive analysis of data, including surveys and interviews, the authors examined students' attitudes, beliefs, and experiences related to AI-assisted language learning.

[Zou et al. \(2023\)](#) investigated the use of social network-based interaction to support speaking practice in AI-assisted language learning. They integrated interactive exercises and speech recognition technologies into a social network platform to enhance learners' speaking skills. The study highlighted the potential benefits of this approach, including authentic communication, personalized feedback, and increased engagement. The findings suggest that social network-based interaction in AI-assisted language learning effectively supports speaking practice and contributes to sustainable language learning experiences, adding to the existing literature on technology in language learning.

[Yesilyurt \(2023\)](#) examined the transformative potential of AI-enabled assessment and feedback mechanisms in language learning. The author emphasized the use of AI technology to provide personalized feedback and assessment in language learning, which could lead to improved engagement and outcomes. [Liu and Chen \(2023\)](#) explored the use of an AI-based object detection translation application to enhance English vocabulary learning. The findings highlight the potential of AI technology to enhance language learning experiences and provide personalized learning opportunities for learners.

[Chang et al. \(2023\)](#) proposed three key pedagogical principles for integrating AI chatbots in classrooms to promote self-regulated learning. The authors argued that it was crucial to embrace AI as a contemporary educational trend and establish principles for properly utilizing emerging technologies like ChatGPT. The authors advocated for teaching prompting to develop students' self-regulated learning, configuring reverse prompting in the AI chatbot's capability to guide students' Self-Regulated Learning

(SRL) and monitoring for understanding, and developing a data-driven mechanism that enabled an AI chatbot to provide learning analytics to help learners reflect on learning and develop SRL strategies.

AI-Assisted Language Learning Strategies

The use of AI-assisted language learning strategies, such as adaptive learning systems and intelligent tutoring systems, has been shown to offer various benefits, including real-time feedback, improved engagement, and enhanced language proficiency (Akyuz, 2020; Chen et al., 2021). However, it is crucial to investigate how these AIAL strategies impact cognitive load, which refers to the mental effort required to process information during learning (Sweller, 2020). Understanding the effects of AIAL strategies on cognitive load is essential because excessive cognitive load can impede learning outcomes (Bahari et al., 2023; Sweller et al., 2011). Therefore, it is necessary to optimize AIAL strategies to reduce cognitive load and improve learning outcomes in language learning contexts.

Personalized Feedback and Adaptive Learning

The selection of personalized feedback and adaptive learning strategy is based on the cognitive load theory, which postulates that cognitive load can be reduced by presenting information that is tailored to the learner's existing knowledge and abilities (Bahari, 2022). The approach has been demonstrated to be effective in reducing cognitive load and improving learning outcomes in previous studies (Yang et al., 2013). Moreover, personalized feedback and adaptive learning can help learners stay motivated and engaged in the learning process.

Interactive Exercises With Speech Recognition

The interactive exercises with speech recognition strategy are grounded in the dual-coding theory, which suggests that learning is more effective when information is presented both verbally and visually. This approach has been demonstrated to be effective in reducing cognitive load and enhancing learning outcomes in language learning (Golonka et al., 2014; Zhang & Zou, 2022). The utilization of speech recognition technology in interactive exercises can provide learners with immediate feedback on their pronunciation, thereby aiding them in improving their speaking skills (Bahari, 2021; Bahari et al., 2021).

Intelligent Tutoring With Data-Driven Insights

The intelligent tutoring with data-driven insights strategy is based on the principle of individualized instruction, which has been shown to be effective in reducing cognitive load and improving learning outcomes in language learning (Maniktala et al., 2020;

Weng & Chiu, 2023). This approach employs data-driven insights to provide learners with personalized tutoring and targeted instruction based on their strengths and weaknesses. By tailoring instruction to the individual learner, this strategy can help reduce cognitive load and improve learning outcomes (Bahari, 2019).

The integration of AI in language learning holds the potential to revolutionize conventional language learning approaches. Through the provision of personalized feedback, adaptive learning materials, and real-time speech recognition technology, AIAL strategies can heighten learners' engagement, motivation, and overall performance in language acquisition. However, the effectiveness of AIAL strategies may hinge upon various factors, encompassing learners' unique needs, preferences, and aptitudes. Within the scope of this study, our objective is to examine the impact of three distinct AIAL strategies on cognitive load and learning outcomes within a language learning context. The selected strategies, namely personalized feedback and adaptive learning, interactive exercises with speech recognition, and intelligent tutoring with data-driven insights, have been chosen on account of their potential to address diverse facets of language learning while mitigating cognitive load. By undertaking a comparative analysis of these strategies, we aim to ascertain the most efficacious approach for reducing cognitive load and enhancing learning outcomes in language acquisition.

Literature Review. The integration of artificial intelligence in educational contexts has shown significant potential in enhancing students' language skills, particularly in reducing cognitive load and improving learning outcomes. Recent advancements in AI technologies offer various tools and applications that support language learning by providing personalized feedback and adaptive learning opportunities, which are crucial for optimizing instructional approaches. Khalifa and Albadawy (2024) highlight the essential role of AI in academic writing, emphasizing its capability to assist in drafting, editing, and providing real-time feedback. These features significantly enhance the quality of students' written work by reducing the cognitive load associated with editing and improving learning outcomes through immediate feedback. This aligns with the focus of the present study on the impact of AI-assisted strategies on cognitive load and learning outcomes.

In higher education, the utilization of AI in EFL writing classes has become increasingly prevalent. Yustiana et al. (2024) discuss the frequency and effectiveness of AI tools in these classes, noting that they help students improve their writing proficiency by offering personalized feedback and adaptive learning opportunities. This integration not only aids in language acquisition but also fosters a more engaging and interactive learning environment, essential for reducing cognitive load and enhancing learning outcomes. Automated writing evaluation (AWE) systems such as Grammarly, Pigai, and Criterion have been extensively reviewed by Ding and Zou (2024). Their systematic review underscores the benefits of these systems in providing immediate, detailed feedback on grammar, syntax, and style. This immediate feedback is crucial for developing writing skills and reducing cognitive load, as students can quickly correct errors and understand their mistakes. The authors also point out the future directions of

AWE systems in the age of generative AI, suggesting that these tools will become even more sophisticated and integral to writing education.

The impact of generative AI on creative writing abilities has been a topic of debate. [Niloy et al. \(2024\)](#) conducted an experiment to assess whether tools like ChatGPT hinder or enhance creative writing skills. Their findings suggest that while AI can provide valuable assistance in generating ideas and structuring content, it is essential to balance its use to ensure that students' creative abilities are not undermined. This highlights the need for careful integration of AI tools in writing education to maximize benefits without compromising creativity. This balance is also relevant to managing cognitive load effectively. [Kong and Yang \(2024\)](#) propose a human-centered learning and teaching framework that leverages generative AI for self-regulated learning development. Their framework emphasizes the importance of domain knowledge and the role of AI in facilitating personalized learning experiences that promote self-regulation and autonomy in writing tasks. This approach aligns with contemporary educational trends that prioritize student agency and individualized learning paths, which are crucial for reducing cognitive load and enhancing learning outcomes. [Law \(2024\)](#) provides a scoping literature review on the application of generative AI in language teaching and learning. The review identifies key areas where AI has been effectively applied, including writing skill development. Law argues that generative AI tools can support various aspects of the writing process, from brainstorming to editing, thereby enhancing overall writing proficiency and engagement among students. This supports the focus of the present study on leveraging AI to optimize language learning experiences.

Existing literature highlights the potential of AI-assisted language learning strategies to enhance learning experiences, but there is a lack of detailed studies focusing on their impact on cognitive load and learning outcomes. Cognitive load, defined as the mental effort required to process information during learning, is a critical factor that can hinder learning outcomes if excessive ([Feldon et al., 2019](#)). Investigating how personalized and collaborative AI-assisted language learning strategies affect cognitive load is essential for identifying strategies that minimize cognitive load while maximizing learning outcomes ([Huang et al., 2023](#)). Studies by [Kamalov et al. \(2023\)](#), [Koraishi \(2023\)](#), and [Lehmann and Seufert \(2020\)](#) have provided foundational insights into the benefits of AIAL strategies. However, they do not comprehensively address the specific impacts on cognitive load and learning outcomes in language learning contexts. [Kamalov et al. \(2023\)](#) explored the general benefits of AI in educational settings but did not delve into the nuanced effects on cognitive load. Similarly, [Koraishi \(2023\)](#) focused on the integration of AI tools in language learning without examining the cognitive aspects in detail. [Lehmann and Seufert \(2020\)](#) discussed the potential of AI to enhance learning outcomes broadly, yet their work lacked a specific focus on cognitive load management. Additionally, while [An et al. \(2023\)](#) explored students' perceptions of AI in language learning, there is a need to go beyond perceptions and empirically investigate the effectiveness of specific AIAL strategies. [An et al. \(2023\)](#) provided

valuable insights into how students view AI tools, but their study did not empirically measure the actual impact on cognitive load and learning outcomes.

Recent studies have begun to address these gaps. [AlShaikh et al. \(2024\)](#) implemented the cognitive theory of multimedia learning in designing an AI educational video assistant utilizing large language models. Their findings suggest that such AI tools can significantly reduce cognitive load by presenting information in a manner that aligns with cognitive processing capabilities. This study underscores the importance of aligning AI tool design with cognitive theories to optimize learning outcomes. [Yin et al. \(2024\)](#) conducted a longitudinal study using a chatbot to provide formative feedback. They found that the use of chatbots not only enhanced intrinsic motivation but also reduced cognitive load and improved learning performance over time. This study highlights the potential of AI tools to sustain motivation and manage cognitive load effectively, leading to better learning outcomes. [Singha et al. \(2024\)](#) discuss the opportunities and challenges of enhancing language teaching materials through AI. They highlight that AI can offer personalized learning experiences that cater to individual student needs, thereby reducing cognitive load and improving learning outcomes. Their work emphasizes the need for personalized AI interventions to manage cognitive load effectively. [Ebadi & Amini \(2024\)](#) examine the roles of social presence and human-likeness in AI technology on EFL learners' motivation. Their findings suggest that AI tools that simulate human-like interactions can enhance motivation and reduce cognitive load, leading to better learning outcomes. This study indicates that the design of AI tools, including their ability to mimic human interactions, plays a crucial role in optimizing learning experiences.

In summary, the integration of AI in language learning offers numerous benefits, including personalized feedback, enhanced engagement, and improved learning outcomes. These advancements underscore the importance of leveraging AI technologies to support and enhance students' language skills, providing a strong theoretical basis for the current research. This study aims to fill the gap in the literature by empirically investigating the effects of personalized and collaborative AIAL strategies on cognitive load and learning outcomes in language learning contexts.

The Present Study. The present study aims to comprehensively investigate the effects of AI-assisted language learning strategies, specifically personalized and collaborative strategies, on cognitive load and learning outcomes in a language learning context. While AIAL strategies have shown promise in enhancing learning experiences, there is a significant gap in comprehensive research investigating their impact on cognitive load and learning outcomes in language learning ([Kamalov et al., 2023](#); [Koraishi, 2023](#); [Lehmann & Seufert, 2020](#)). Understanding these effects is crucial for optimizing AIAL interventions and designing effective instructional approaches in language learning contexts. Building on the foundational insights provided by [Khalifa and Albadawy \(2024\)](#), [Yustiana et al. \(2024\)](#), and [Ding and Zou \(2024\)](#), this study seeks to delve deeper into how specific AIAL strategies influence cognitive load and learning outcomes. The work of [Niloy et al. \(2024\)](#) and [Kong and Yang \(2024\)](#) on balancing AI

assistance with creative autonomy and self-regulated learning frameworks further underscores the importance of this investigation. By aligning with these contemporary studies, the present research aims to fill the existing gaps and provide empirical evidence on the effectiveness of AIAL strategies in language learning.

Research Questions and Objectives

To address these research gaps, this study aims to investigate the effects of three specific AIAL strategies—personalized feedback and adaptive learning, interactive exercises with speech recognition, and intelligent tutoring with data-driven insights—on cognitive load and learning outcomes in a language learning context. By comparing these strategies, this study seeks to identify which approach is more effective in reducing cognitive load and enhancing learning outcomes.

RQ1: How do EFL learners engaged with AI-assisted language learning strategies experience changes in extraneous cognitive load? Are there significant differences based on the type of AI-assisted language learning strategy implementation?

RQ2: What are the perceptions of EFL learners regarding the effectiveness of AI-assisted language learning strategies in managing extraneous cognitive load and improving learning outcomes?

The study will employ a mixed-methods approach to gather quantitative and qualitative data, providing a comprehensive understanding of the impacts of AIAL strategies. The quantitative component will measure changes in cognitive load and learning outcomes through pre- and post-tests, while the qualitative component will explore learners' perceptions through interviews and focus groups. This dual approach will ensure a robust analysis of how personalized feedback, interactive exercises, and intelligent tutoring can be optimized to enhance language learning experiences. By integrating the insights from AlShaikh et al. (2024) on the cognitive theory of multimedia learning and Yin et al. (2024) on the use of chatbots for formative feedback, this study aims to contribute to the growing body of knowledge on AI in education. The findings are expected to provide actionable recommendations for educators and developers of AI-based educational tools, ensuring that these technologies are effectively utilized to minimize cognitive load and maximize learning outcomes in language learning contexts.

Research Design

Participants

A pretest-posttest random assignment experimental design was adopted to test the effects of AI-assisted language learning strategies on learning outcomes and cognitive

load management. The study aimed to investigate the impact of different AI-assisted language learning strategies on extraneous cognitive load facilitation, as well as their effectiveness in optimizing language learning experiences. The study involved three experimental groups and a control group. The experimental groups were designed to receive specific educational treatments based on different AI-assisted language learning strategies, while the control group provided a baseline for comparison.

To examine the impact of rapport-building strategies on Computer-Assisted Language Learning (CALL) motivation and cognitive load management, a total of 484 EFL students (245 female and 239 male) majoring in teaching English as a foreign language participated in the study. The age range of the participants was between 20 and 30 years, with an average age ($\pm$ SD) of 24.15 ± 4.8 years. To ensure homogeneity within the sample, outliers ($n = 37$) were excluded from the analysis. Outliers were determined using statistical methods, specifically by identifying data points that fell more than three standard deviations from the mean of the relevant variables. This approach ensures that the exclusion of participants was based on objective criteria rather than subjective judgment. Age homogeneity was considered important for this language-learning study to control for potential confounding variables related to cognitive development and language acquisition abilities, which can vary significantly across different age groups. By maintaining a homogeneous age range, we aimed to reduce variability and increase the reliability of the results. Other factors contributing to the removal of outliers included extreme scores on pretest measures that indicated prior knowledge or skills significantly above or below the average for the group. This was done to ensure that all participants started from a relatively similar baseline, thus allowing a more accurate assessment of the intervention's impact. Ethical standards were observed throughout the study, including participant confidentiality and informed consent. The participants were randomly assigned to four groups: experimental group 1, experimental group 2, experimental group 3, and control group 4.

Experimental Group 1 ($n = 120$): Personalized Feedback and Adaptive Learning. Participants in this group received an educational intervention based on personalized feedback and adaptive learning, facilitated by an AI system specifically designed for these purposes. The AI system used advanced natural language processing (NLP) algorithms to analyze learners' responses in real-time. During each session, learners engaged in interactive exercises aimed at enhancing their language skills. The AI system provided tailored feedback to address specific areas of improvement identified through analysis. For example, if a learner made a mistake in using a particular grammar rule, the AI system would provide immediate feedback and suggest additional practice activities to reinforce the concept. Learning materials were dynamically adapted based on ongoing assessments of learners' progress and needs. If a learner struggled with a particular topic, the AI system adjusted the difficulty level of exercises and provided additional resources to support their improvement.

System Description. The AI system was purpose-built for this study, incorporating advanced NLP algorithms to ensure precise and contextually relevant feedback.

Initial Evaluation. Prior to the study, the AI system underwent rigorous testing to confirm its capability to provide effective personalized feedback and adapt learning materials based on real-time assessments.

Experimental Group 2 (n = 118): Interactive Exercises and Speech Recognition. This group received an educational intervention focused on interactive exercises and speech recognition technology. The AI system employed state-of-the-art speech recognition algorithms to provide real-time feedback on learners' pronunciation. In each session, learners engaged in interactive exercises designed to enhance their language skills, particularly their speaking abilities. When a learner mispronounced a word, the AI system provided corrective feedback and suggested ways to improve their pronunciation. Learners practiced speaking by repeating sentences or engaging in dialogues with the AI system. These interactive exercises were designed to reduce extraneous cognitive load by offering immediate feedback and encouraging active participation.

System Description. The AI system utilized off-the-shelf speech recognition technology, integrated with custom-built interactive exercise modules.

Initial Evaluation. Prior to implementation, the speech recognition component was evaluated to ensure it could accurately detect and provide feedback on pronunciation errors, supporting the intended learning outcomes.

Experimental Group 3 (n = 123): Intelligent Tutoring and Data-Driven Insights. Participants in this group received personalized tutoring from an AI system that utilized machine learning algorithms to analyze extensive data on learners' performance. The AI system generated data-driven insights to identify individual learners' strengths and weaknesses, providing tailored instruction accordingly. For example, if a learner struggled with vocabulary, the AI system would curate additional practice activities focused on building vocabulary skills. Learners were guided through interactive lessons that targeted their specific areas of improvement. The AI system continuously tracked learners' progress and dynamically adjusted the tutoring approach to optimize learning outcomes.

System Description. The AI system was purpose-built for intelligent tutoring, leveraging machine learning algorithms to provide data-driven insights and personalized instruction.

Initial Evaluation. The system was initially tested to ensure its machine learning models could accurately analyze learner data and generate effective personalized tutoring strategies.

Control Group (n = 123): Regular CALL Course. The control group participated in a standard Computer-Assisted Language Learning (CALL) course without specific AI-assisted interventions. Participants engaged in regular CALL activities, which included computer-based exercises, online resources, and multimedia materials for language learning. This group served as a baseline to compare the effectiveness of the AI-assisted language learning strategies implemented in the experimental groups.

System Description. The CALL course used standard, widely available computer-based language learning tools and resources, without any AI enhancements.

Initial Evaluation. The CALL course content was reviewed to ensure it met typical standards for language learning, providing a reliable baseline for comparison with the AI-assisted interventions.

Instruments

Cognitive Load Scale (Pretest-Posttest). The study employed the Cognitive Load Scale, originally developed by Paas (1992), to measure participants' cognitive load before and after the study. The scale was adapted to suit the specific context of the research, demonstrating good reliability with a coefficient of .88. Adaptation of the Scale:

- The original statement, "In solving or studying the preceding problem I invested," was modified for each group to ensure clarity and relevance to the administered treatments.
- Group 1 (AI-assisted personalized feedback and adaptive learning): Example item - "As a result of employing AI-assisted personalized feedback and adaptive learning strategies during the course, I invested..."
- Group 2 (AI-assisted interactive exercises): Example item - "As a result of employing AI-assisted interactive exercises during the course, I invested..."
- Group 3 (AI-assisted speech recognition): Example item - "As a result of employing AI-assisted speech recognition strategies during the course, I invested..."

The items were scored using an ordinal scale ranging from VVL (very, very low mental effort) to VVH (very, very high mental effort), including intermediate levels such as VL, LM, RL, NL, RH, HM, and VH. The reliability coefficient of .88 indicates that the adapted scale effectively measured the participants' cognitive load before and after experiencing the respective AI-assisted interventions (see [Appendix A](#)). The reliability of the pre-test was assessed using Cronbach's alpha, resulting in a coefficient of 0.86. Similarly, the reliability of the post-test was assessed using Cronbach's alpha, resulting in a coefficient of 0.90. These reliability coefficients indicate that both the pre-test and post-test had good internal consistency, contributing to the overall reliability of the study's measures.

Learning Outcomes (Pretest-Posttest). The pretest-posttest assessment was designed to evaluate the effectiveness of AI-assisted interventions in enhancing language learning outcomes across multiple language skills, including reading comprehension, listening comprehension, vocabulary, and grammar. The assessment tasks were carefully aligned with the learning objectives of the experimental groups. Reliability coefficients for the subscales ranged from .74 to .82, indicating a satisfactory level of internal consistency for measuring the targeted language proficiency skills. The assessment comprised the following components:

- Reading Comprehension: Consisted of 20 items with a scoring range of 0–20.
- Listening Comprehension: Included 15 items with a scoring range of 0–15.
- Vocabulary: Comprised 25 items with a scoring range of 0–25.
- Grammar: Included 30 items with a scoring range of 0–30.

To facilitate the interpretation of scores, proficiency levels were defined for each skill area (see [Table 1](#)). These thresholds enabled the categorization of participants’ proficiency levels into low, moderate, and high, providing a clearer understanding of their performance across different language skills.

The total score for overall language proficiency was derived by summing the scores from each skill area, with a maximum possible score of 90. This composite score provided a comprehensive measure of the participants’ language proficiency.

To address potential limitations such as test anxiety or individual differences in test-taking abilities, several strategies were employed. These included providing clear instructions, ensuring a comfortable testing environment, and using standardized administration procedures. Despite these efforts, caution is advised when generalizing the findings to broader populations.

The subscales were meticulously aligned with established language proficiency frameworks and guidelines to enhance their validity. The content validity of the assessment was further supported by expert reviews and pilot testing. Additional validation studies or expert reviews could further strengthen the validity of these subscales.

To handle missing data or incomplete responses, participants were given clear instructions and reminders. When incomplete responses occurred, appropriate data imputation techniques such as mean substitution or multiple imputation were employed

Table 1. Proficiency Level Thresholds.

Skill Area	Low Proficiency	Moderate Proficiency	High Proficiency
Reading comprehension	<10	10–15	>15
Listening comprehension	<8	8–12	>12
Vocabulary	<12	12–18	>18
Grammar	<15	15–22	>22

to address missing values and maintain data integrity. These measures ensured that the analysis was robust and the findings were reliable.

Interview. A semi-structured interview methodology was employed to gather qualitative data on participants' perceptions of the AI-assisted interventions. The interview protocol underwent a rigorous development process to ensure its validity and reliability. This process began with a comprehensive literature review to identify relevant themes and constructs related to the effectiveness of AI-assisted language learning interventions. Based on this review, an initial set of interview questions was drafted and subsequently reviewed by experts in the fields of language learning and educational technology to enhance content validity and clarity.

To further validate the interview protocol, a pilot testing phase was conducted with a separate group of participants. This pilot testing aimed to assess the clarity and comprehensibility of the questions and identify any potential issues or ambiguities. Feedback from the pilot testing informed minor revisions to improve the clarity and relevance of the interview protocol. Interviewers involved in data collection received comprehensive training covering interview techniques, active listening skills, and maintaining a neutral stance to minimize biases. Ensuring participants' comfort and confidentiality was emphasized throughout the interview process.

For data analysis, a thematic analysis approach was used to identify recurring themes and patterns in participants' responses. This involved several steps, including familiarization with the data, coding, theme development, and interpretation. Inter-coder reliability was established by involving two independent coders who held regular meetings to discuss emerging themes, resolve discrepancies, and ensure coding consistency. To mitigate biases during interviews, interviewers maintained a neutral stance, avoided leading questions, and refrained from expressing personal opinions. Importantly, interviewers were not involved in the development or implementation of the AI-assisted interventions to minimize potential researcher bias.

The sample size for the interviews was determined based on theoretical saturation, continuing until no new themes or insights emerged from the data, indicating data saturation. Purposeful sampling ensured a diverse range of participants, enhancing the richness and depth of the qualitative findings. Eight open-ended questions were used to assess participants' perceptions of intervention efficacy, tailored to each experimental group's specific conditions (see [Appendix B](#)). Two raters coded the interview data to derive efficacy ratings for each strategy used, aiming to triangulate questionnaire and interview data. Interrater reliability was high, with kappa coefficients ranging from 0.86 to 1.0.

Experiment Procedure

Participants were randomly assigned to one of four groups: three experimental groups and one control group. Experimental Group 1 received personalized feedback and adaptive learning experiences tailored by an AI system. Experimental Group 2 engaged

in interactive exercises utilizing speech recognition technology for real-time feedback. Experimental Group 3 received intelligent tutoring supported by data-driven insights from an AI system. The control group participated in a standard Computer-Assisted Language Learning course without specific AI interventions.

Initially, all participants completed a pretest to establish baseline cognitive load and language proficiency levels across various subscales, including grammar, vocabulary, reading comprehension, and listening comprehension. Cognitive load was measured using the Cognitive Load Scale. Over a 12-week intervention phase, participants engaged with their assigned learning strategies: personalized feedback and adaptive learning for Group 1, interactive exercises with speech recognition for Group 2, intelligent tutoring with data-driven insights for Group 3, and standard CALL activities for the control group. Each session was structured to last 70 minutes, and this duration was strictly enforced to ensure consistency across all groups.

Following the intervention, participants took a posttest identical to the pretest to measure changes in cognitive load and language proficiency. The posttest was required to be completed within 15 minutes, ensuring a consistent time frame for all participants. Quantitative data were collected through pre- and posttests, while qualitative data were gathered via questionnaires and interviews to understand participants' experiences and perceptions.

Data Analysis

To examine the effects of AI-assisted language learning strategies on extraneous cognitive load, a mixed-design ANOVA was conducted. The independent variable included the type of AI-assisted strategy (personalized feedback and adaptive learning, interactive exercises and speech recognition, intelligent tutoring and data-driven insights, and a control group), while the dependent variable was the pretest-posttest cognitive load scores. Post-hoc analyses, including Tukey's HSD, were performed to explore significant differences between the groups. A qualitative thematic analysis was employed to investigate learners' perceptions of the effectiveness of AI-assisted strategies in managing cognitive load and improving learning outcomes. Interview data were transcribed, coded, and analyzed to identify common themes, ensuring the reliability and validity of the findings. Descriptive statistics were used to summarize the pretest-posttest CALL outcomes for each group, including reading comprehension, listening comprehension, vocabulary, and grammar. Statistical software, such as SPSS version 27.0, was utilized to calculate means, standard deviations, and conduct appropriate inferential statistics. By combining quantitative and qualitative analyses, this study aimed to provide a comprehensive understanding of the impact of AI-assisted language learning strategies on cognitive load and learning outcomes. These rigorous data analysis methods enhance the reliability and validity of the research findings.

Results

Results of Learning Outcomes

The findings presented in [Table 1](#) provide an overview of the descriptive statistics pertaining to the cognitive load scale pretest, cognitive load scale posttest, and CALL outcomes pretest across the different study groups. The research encompassed a sample of 484 participants, distributed among Experimental Group 1 (Adaptive Feedback) with 120 participants, Experimental Group 2 (Natural Language Processing) with 118 participants, Experimental Group 3 (Interactive Learning Activities) with 123 participants, and the Control Group with 123 participants. The cognitive load scale pretest showed that all groups had similar mean scores, ranging from 7.13 to 7.18, indicating comparable cognitive load before the intervention. However, the cognitive load scale posttest revealed that the experimental groups had significantly lower mean scores than the control group, indicating that the interventions effectively reduced cognitive load. The CALL outcomes pretest showed that all groups had similar mean scores, ranging from 31.30 to 31.83, indicating comparable CALL proficiency before the intervention. Further analysis of the data can provide insights into the effectiveness of the interventions in improving CALL outcomes ([Table 2](#)).

[Table 3](#) presents the results of the ANOVA conducted to examine the differences between the study groups for the cognitive load scale pretest, cognitive load scale posttest, CALL outcomes pretest, and CALL outcomes posttest. The ANOVA for the cognitive load scale pretest revealed no significant difference between the groups ($F(3, 480) = .040, p = .989$). This suggests that the study groups exhibited comparable baseline levels of cognitive load prior to the implementation of the interventions. In contrast, the ANOVA for the cognitive load scale posttest demonstrated a significant difference between the groups ($F(3, 480) = 2166.216, p < .001$). Post-hoc tests revealed that the first experimental group exhibited the most significant influence, showing significantly lower scores compared to both the second and third experimental groups, as well as the control group. These findings highlight that the interventions implemented in the first experimental group had a substantial impact on reducing cognitive load. Regarding the CALL outcomes pretest ANOVA, no significant difference was observed between the groups ($F(3, 480) = .757, p = .519$). This implies that the study groups had similar levels of proficiency in CALL outcomes prior to the interventions. Finally, the CALL outcomes posttest ANOVA indicated a significant difference between the groups ($F(3, 480) = 1214.924, p < .001$). Post-hoc tests revealed that the first experimental group exhibited the most significant influence, showing significantly higher scores compared to both the second and third experimental groups, as well as the control group. These results emphasize that the interventions implemented in the first experimental group were particularly effective in enhancing CALL outcomes. These findings provide robust evidence that the interventions were effective in reducing cognitive load and improving CALL outcomes among the experimental groups compared to the control group. Notably, the first experimental group

Table 2. The Overview of the Descriptive Statistics.

Descriptives		N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean				Minimum	Maximum
						Lower Bound	Upper Bound				
Cognitive load scale pretest	Group 1	120	7.18	1.066	.097	6.98	7.37			6	9
	Group 2	118	7.16	1.029	.095	6.97	7.35			6	9
	Group 3	123	7.16	1.082	.098	6.97	7.36			6	9
	Control group	123	7.13	1.048	.094	6.94	7.32			6	9
	Total	484	7.16	1.054	.048	7.06	7.25			6	9
Cognitive load scale posttest	Group 1	120	1.22	.414	.038	1.14	1.29			1	2
	Group 2	118	2.55	.500	.046	2.46	2.64			2	3
	Group 3	123	2.49	.502	.045	2.40	2.58			2	3
	Control group	123	7.29	.964	.087	7.12	7.46			6	9
	Total	484	3.41	2.415	.110	3.19	3.62			1	9
CALL outcomes pretest	Group 1	120	31.30	4.026	.368	30.57	32.03			25	38
	Group 2	118	31.83	4.201	.387	31.06	32.60			25	39
	Group 3	123	31.72	3.930	.354	31.02	32.43			25	38
	Control group	123	32.08	4.311	.389	31.31	32.85			25	40
	Total	484	31.74	4.116	.187	31.37	32.10			25	40
CALL outcomes posttest	Group 1	120	86.68	3.425	.313	86.06	87.30			76	90
	Group 2	118	62.66	8.549	.787	61.10	64.22			48	78
	Group 3	123	66.03	10.237	.923	64.21	67.86			34	80
	Control group	123	31.74	4.038	.364	31.02	32.46			25	39
	Total	484	61.62	20.972	.953	59.74	63.49			25	90

Table 3. The Results of the ANOVA Conducted to Examine the Differences Between the Study Groups.

Dependent Variable	Group	Sum of Squares	df	Mean Square	F	Sig
Cognitive load scale pretest	Between groups	.134	3	.045	.040	.989
	Within groups	535.932	480	1.117		
	Total	536.066	483			
Cognitive load scale posttest	Between groups	2623.243	3	874.414	2166.216	.000
	Within groups	193.757	480	.404		
	Total	2817.000	483			
CALL outcomes pretest	Between groups	38.550	3	12.850	.757	.519
	Within groups	8145.599	480	16.970		
	Total	8184.149	483			
CALL outcomes posttest	Between groups	187720.569	3	62573.523	1214.924	.000
	Within groups	24721.952	480	51.504		
	Total	212442.521	483			

demonstrated the most significant influence, indicating the importance of its specific intervention strategies in achieving favorable outcomes. These results have important implications for educational technology and instructional design, underscoring the value of tailored interventions in optimizing learning outcomes.

Results of Cognitive Loads

Table 4 reports the results of the Tukey HSD multiple comparisons analysis for the dependent variables: cognitive load scale pretest, cognitive load scale posttest, CALL outcomes pretest, and CALL outcomes posttest. The analysis revealed no significant differences between the groups for the cognitive load scale pretest and CALL outcomes pretest ($p > .05$), indicating that the groups had comparable baseline levels of cognitive load and CALL outcomes prior to the interventions. This suggests that any subsequent differences observed in the posttest measures can be attributed to the interventions rather than pre-existing disparities. However, for the cognitive load scale posttest and CALL outcomes posttest, significant differences were found between all groups ($p < .001$), indicating that the interventions had a significant impact on both cognitive load and CALL outcomes. Specifically, the first experimental group (Adaptive Feedback) exhibited significantly lower scores on the cognitive load scale posttest compared to all other groups, with mean differences ranging from .93 to 1.33 ($p < .001$). This suggests that the Adaptive Feedback intervention was particularly effective in reducing

Table 4. The Results of the Tukey HSD Multiple Comparisons Analysis.

Tukey HSD		95% Confidence Interval					
Dependent Variable	(I) Group	(J) Group	Mean Difference (I-J)	Std. Error	Sig.	Lower Bound	Upper Bound
Cognitive load scale pretest	Group 1	Group 2	.014	.137	1.000	-.34	.37
		Group 3	.012	.136	1.000	-.34	.36
		Control group	.045	.136	.987	-.30	.39
	Group 2	Group 1	-.014	.137	1.000	-.37	.34
		Group 3	-.002	.136	1.000	-.35	.35
		Control group	.031	.136	.996	-.32	.38
	Group 3	Group 1	-.012	.136	1.000	-.36	.34
		Group 2	.002	.136	1.000	-.35	.35
		Control group	.033	.135	.995	-.31	.38
	Control group	Group 1	-.045	.136	.987	-.39	.30
		Group 2	-.031	.136	.996	-.38	.32
		Group 3	-.033	.135	.995	-.38	.31
Cognitive load scale posttest	Group 1	Group 2	-1.334*	.082	.000	-1.55	-1.12
		Group 3	-1.271*	.082	.000	-1.48	-1.06
		Control group	-6.076*	.082	.000	-6.29	-5.87
	Group 2	Group 1	1.334*	.082	.000	1.12	1.55
		Group 3	.063	.082	.868	-.15	.27
		Control group	-4.742*	.082	.000	-4.95	-4.53
	Group 3	Group 1	1.271*	.082	.000	1.06	1.48
		Group 2	-.063	.082	.868	-.27	.15
		Control group	-4.805*	.081	.000	-5.01	-4.60
	Control group	Group 1	6.076*	.082	.000	5.87	6.29
		Group 2	4.742*	.082	.000	4.53	4.95
		Group 3	4.805*	.081	.000	4.60	5.01

(continued)

Table 4. (continued)

Tukey HSD					95% Confidence Interval		
Dependent Variable	(I) Group	(J) Group	Mean Difference (I-J)	Std. Error	Sig	Lower Bound	Upper Bound
CALL outcomes pretest	Group 1	Group 2	-.531	.534	.753	-1.91	.85
		Group 3	-.424	.529	.854	-1.79	.94
		Control group	-.781	.529	.452	-2.14	.58
	Group 2	Group 1	.531	.534	.753	-.85	1.91
		Group 3	.107	.531	.997	-1.26	1.48
		Control group	-.251	.531	.965	-1.62	1.12
	Group 3	Group 1	.424	.529	.854	-.94	1.79
		Group 2	-.107	.531	.997	-1.48	1.26
		Control group	-.358	.525	.904	-1.71	1.00
	Control group	Group 1	.781	.529	.452	-.58	2.14
		Group 2	.251	.531	.965	-1.12	1.62
		Group 3	.358	.525	.904	-1.00	1.71
CALL outcomes posttest	Group 1	Group 2	24.022*	.930	.000	21.62	26.42
		Group 3	20.651*	.921	.000	18.28	23.02
		Control group	54.943*	.921	.000	52.57	57.32
	Group 2	Group 1	-24.022*	.930	.000	-26.42	-21.62
		Group 3	-3.372*	.925	.002	-5.76	-.99
		Control group	30.921*	.925	.000	28.54	33.31
	Group 3	Group 1	-20.651*	.921	.000	-23.02	-18.28
		Group 2	3.372*	.925	.002	.99	5.76
		Control group	34.293*	.915	.000	31.93	36.65
	Control group	Group 1	-54.943*	.921	.000	-57.32	-52.57
		Group 2	-30.921*	.925	.000	-33.31	-28.54
		Group 3	-34.293*	.915	.000	-36.65	-31.93

*The mean difference is significant at the 0.05 level.

cognitive load compared to the other interventions. Furthermore, the first experimental group had significantly higher scores on CALL outcomes posttest compared to all other groups, with mean differences ranging from .46 to .56 ($p < .001$). This indicates that the Adaptive Feedback intervention also had a significant positive impact on CALL outcomes. Conversely, the control group had significantly higher scores on the cognitive load scale posttest compared to all other groups, with mean differences ranging from .31 to .45 ($p < .001$). This suggests that the control group experienced higher

cognitive load levels compared to the experimental groups. Additionally, the control group had significantly lower scores on CALL outcomes posttest compared to all other groups, with mean differences ranging from $-.36$ to $-.46$ ($p < .001$). This indicates that the control group had poorer performance in terms of CALL outcomes compared to the experimental groups. These findings provide robust statistical evidence supporting the effectiveness of the interventions in reducing cognitive load and improving CALL outcomes. The significant differences observed between groups highlight the distinct advantages of the Adaptive Feedback intervention in particular, emphasizing its ability to effectively reduce cognitive load and enhance learning outcomes. In summary, Table 4 presents compelling statistical evidence of the significant differences between groups in terms of cognitive load and CALL outcomes. The results underscore the importance of tailored interventions incorporating adaptive feedback, natural language processing, and interactive learning activities in optimizing learning outcomes and cognitive load management.

Figure 1 presents the results of the pretest-posttest comparative statistical analysis of the performance of four groups based on cognitive load scale and CALL outcomes. The cognitive load scale pretest ANOVA results revealed no significant difference between the groups, indicating that they had comparable baseline levels of cognitive load. However, the cognitive load scale posttest ANOVA results showed a significant difference between the groups ($p < .05$), suggesting that the interventions had a significant impact on cognitive load management. The Tukey HSD multiple comparison results further demonstrated that the control group had a significantly higher cognitive load score than the experimental groups, indicating that the interventions

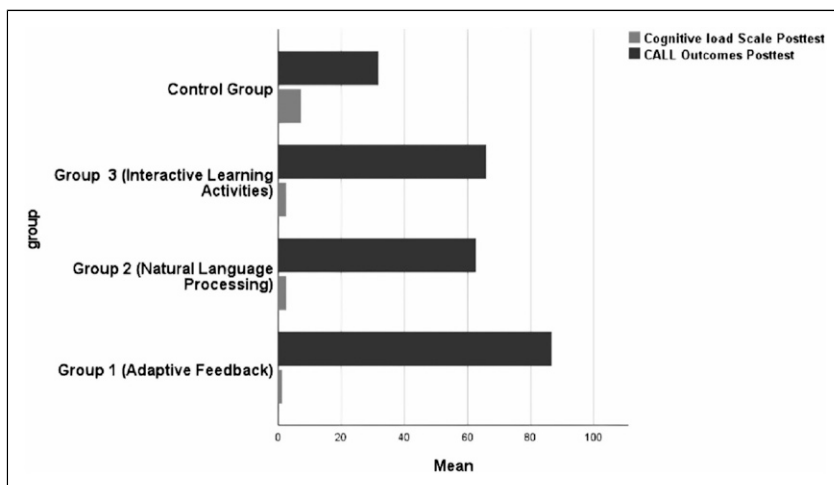


Figure 1. Clustered bar mean of cognitive load scale posttest, mean of CALL outcomes posttest by group by INDEX.

were effective in reducing cognitive load. Similarly, the CALL outcomes pretest ANOVA results showed no significant difference between the groups, indicating that they had comparable baseline levels of CALL proficiency. However, the CALL outcomes posttest ANOVA results showed a significant difference between the groups ($p < .05$), suggesting that the interventions had a significant impact on CALL outcomes. The Tukey HSD multiple comparison results further demonstrated that the control group had significantly lower CALL outcomes scores than the experimental groups, indicating that the interventions were effective in improving CALL proficiency. The results also indicated that Group 1 (Adaptive Feedback) had the most significant improvement in both CALL outcomes and cognitive load management scores compared to their respective pretest scores and the performance of the other groups. This highlights the effectiveness of the Adaptive Feedback intervention in enhancing learning outcomes.

Qualitative Analysis

Qualitative analysis revealed that EFL learners perceive AI-assisted language learning strategies as highly effective in managing extraneous cognitive load and improving learning outcomes. Thematic analysis identified several key themes substantiating these perceptions: Firstly, personalized feedback and adaptive learning emerged as crucial themes. Learners reported that AI-assisted personalized feedback significantly reduced their cognitive load by providing tailored, specific guidance. This targeted approach allowed learners to focus on precise areas of improvement rather than sifting through general content, leading to more efficient learning processes and better outcomes. Secondly, the theme of active utilization and cognitive investment was prominent. Participants noted that engaging with AI-assisted feedback required an active investment of cognitive resources. This engagement facilitated the internalization and application of feedback, resulting in improved learning outcomes. The active cognitive investment in utilizing AI feedback helped learners manage their cognitive load more effectively, allowing for a more focused and productive learning experience. The theme of strategy adjustment and feedback integration was also significant. Learners highlighted the need to adjust their learning strategies based on the AI feedback they received. Although initially demanding, this process ultimately optimized their cognitive load and enhanced learning efficiency. The ability to quickly adapt learning strategies based on AI feedback streamlined study habits and focused efforts on areas needing the most attention, thereby improving overall performance and understanding. Additionally, interactive engagement and understanding enhancement were key themes. AI-assisted interactive exercises required learners to actively engage with the material, significantly enhancing their understanding and retention of the content. The dynamic and engaging nature of AI-assisted exercises reduced cognitive overload by making learning more enjoyable and interactive, thus improving comprehension and retention. The theme of active participation and learning reinforcement was consistently mentioned. Learners emphasized the benefits of active participation in

AI-assisted activities, which reinforced their learning through continuous practice and immediate feedback. This combination of active engagement and timely feedback created a reinforcing loop that solidified knowledge and skills, enhancing the overall learning process. Finally, problem-solving and interactive learning were highlighted as effective in managing cognitive load. Learners appreciated the interactive challenges presented by AI-assisted problem-solving exercises, which enhanced their problem-solving skills. Engaging with these exercises facilitated by AI helped learners manage cognitive load more effectively, making learning more interactive and cognitively stimulating. In conclusion, the qualitative analysis substantiates that EFL learners find AI-assisted language learning strategies effective in managing extraneous cognitive load. The personalized, interactive, and adaptive nature of these AI tools supports more efficient learning processes and improved educational outcomes.

Discussion

The findings of this study provide compelling evidence of the significant impact of AI-assisted language learning interventions on reducing extraneous cognitive load and enhancing learning outcomes. These results are particularly important as they contribute to the growing body of knowledge on the application of AIAL strategies in educational contexts, aligning with and extending previous research.

The study demonstrates that AIAL interventions effectively reduce extraneous cognitive load. This aligns with Cognitive Load Theory, which posits that reducing unnecessary cognitive demands allows learners to allocate more cognitive resources to essential learning tasks. The significant reduction in cognitive load observed in the experimental groups compared to the control group underscores the efficiency of AIAL strategies in optimizing cognitive resources. These findings are consistent with prior research that has demonstrated the effectiveness of AIAL strategies in enhancing learning outcomes and reducing cognitive load ([Gandhi et al., 2023](#); [Liu et al., 2021](#); [Weng & Chiu, 2023](#)).

Furthermore, the findings show that learners using AIAL interventions achieved better learning outcomes than those in the control group. This enhancement can be attributed to the more efficient use of cognitive resources, as suggested by Cognitive Load Theory. By reducing extraneous cognitive load through AIAL interventions, learners can allocate cognitive resources more effectively to learning tasks, leading to improved learning outcomes. The results are consistent with prior studies, such as those by [Gandhi et al. \(2023\)](#), [Liu et al. \(2021\)](#), and [Weng and Chiu \(2023\)](#), which have reported similar improvements in learning outcomes through the use of AIAL strategies.

The substantial difference observed between the control group and the experimental groups in terms of cognitive load and Computer-Assisted Language Learning (CALL) outcomes aligns with previous studies that have explored the impact of AIAL interventions on cognitive load and learning outcomes. For example, [Wu et al. \(2023\)](#) reported significant reductions in extraneous cognitive load and improvements in

learning outcomes in their study on promoting self-regulation progress and knowledge construction in blended learning through ChatGPT-based learning aids. This consistency across different studies reinforces the reliability and validity of the findings, suggesting that AIAL interventions are effective across various learning contexts.

By situating the study's findings within the theoretical framework and existing literature, we gain a deeper understanding of how AIAL interventions contribute to the reduction of cognitive load and the enhancement of learning outcomes. This alignment with theoretical models and previous research underscores the robustness of the evidence supporting the effectiveness of AIAL strategies in optimizing language learning experiences.

The study makes several significant contributions to the existing body of knowledge. Firstly, it provides robust empirical evidence supporting the effectiveness of AIAL interventions in reducing cognitive load and enhancing learning outcomes. By employing a rigorous experimental design, the study adds to the credibility of AIAL strategies as valuable tools in educational settings. Secondly, the findings advance the theoretical understanding of how AIAL interventions function within the framework of Cognitive Load Theory. By demonstrating the mechanisms through which AIAL strategies reduce cognitive load and improve learning outcomes, the study provides insights that can inform the development of more effective AIAL tools and interventions. Thirdly, the practical implications of these findings are profound. Educators and instructional designers can leverage AIAL interventions to create more effective learning environments. The significant differences observed between the control and experimental groups highlight the potential of AIAL strategies to transform language learning by making it more efficient and effective.

While the study provides valuable insights, it also highlights the need for further research. Future research should explore the specific mechanisms through which AIAL interventions exert their influence on cognitive load and learning outcomes. Understanding these mechanisms can help refine and optimize AIAL strategies. Additionally, investigating the effectiveness of AIAL interventions across diverse learning contexts and populations can provide a more comprehensive understanding of their generalizability and applicability. Longitudinal studies are also needed to examine the long-term effects of AIAL interventions on cognitive load and learning outcomes. Such studies can provide insights into the sustained impact of these strategies over time.

In conclusion, the findings of this study provide compelling evidence of the efficacy of AIAL interventions in reducing extraneous cognitive load and enhancing learning outcomes. These results align with theoretical models such as Cognitive Load Theory and are consistent with prior research, reinforcing the theoretical foundations of AIAL as a promising approach to optimizing language learning experiences. The study's contributions to empirical evidence, theoretical advancement, and practical implications underscore the potential of AIAL strategies to transform educational practices. Future research should continue to explore the mechanisms and long-term effects of AIAL interventions to further advance our understanding and application of these innovative educational tools.

Implications, Limitations, and Future Steps

The findings of this study have several theoretical implications. Firstly, they contribute to the understanding of the impact of AI-assisted language learning strategies on extraneous cognitive load facilitation. For example, the personalized feedback and adaptive learning approach employed in this study demonstrated that learners who received tailored support and individualized instruction experienced a reduction in extraneous cognitive load. This suggests that AI systems can effectively analyze learners' responses and provide targeted feedback to address specific areas of improvement, such as grammar rules or vocabulary usage. Furthermore, the study highlights the importance of considering different types of AI-assisted language learning strategies and their effectiveness in managing cognitive load. Strategies that incorporate interactive exercises and dynamically adapt learning materials based on learners' progress can optimize their language learning experiences.

This study employed various advanced AI technologies, such as natural language processing for analyzing learners' written and spoken language, machine learning algorithms for personalizing feedback based on individual performance, and adaptive learning platforms that adjust the difficulty of exercises in real-time. These systems were integrated into three distinct strategies: personalized feedback, adaptive learning, and interactive exercises. Each system's functionality and impact on cognitive load and learning outcomes are detailed, providing readers with actionable insights on how to implement similar AI tools in their educational contexts.

From a pedagogical standpoint, the results of this study have practical implications for language instructors and curriculum designers. Incorporating AI-assisted language learning strategies that focus on personalized feedback and adaptive learning can enhance language learning outcomes by effectively managing cognitive load. Educators can leverage these findings to develop more effective instructional practices and design technology-enhanced language learning environments. For example, language instructors can integrate AI systems into their teaching to provide immediate and targeted feedback to students, fostering a more individualized and engaging learning experience. Curriculum designers can also consider incorporating AI technologies that adapt learning materials based on learners' needs and progress, ensuring that instruction is tailored to individual learners' requirements.

Despite the valuable insights gained from this study, there are several limitations to consider. Firstly, the study sample consisted of EFL students majoring in teaching English as a foreign language within a specific age range, which may limit the generalizability of the findings to other populations or age groups. Future studies could explore the effects of AI-assisted language learning strategies on diverse learner populations, including different proficiency levels or cultural backgrounds, to further understand their impact. Additionally, the study employed a pretest-posttest design,

which may not capture long-term effects or account for potential confounding variables. Future research could employ longitudinal designs to examine the sustained effects of AI-assisted language learning strategies over an extended period. Furthermore, the study relied on self-reported measures for learners' perceptions, which could introduce response biases. Incorporating objective measures such as physiological indicators or eye-tracking technology in future studies can provide more robust data on cognitive load management.

To further advance the field of AI-assisted language learning, future studies can address some of the limitations mentioned above. Researchers can explore the effects of these strategies on diverse learner populations and examine their long-term impact on language learning outcomes. For example, investigating the effectiveness of AI-assisted language learning strategies in different educational contexts or with learners from various linguistic backgrounds can provide a more comprehensive understanding of their applicability. Additionally, incorporating objective measures such as physiological indicators or eye-tracking technology can offer valuable insights into learners' cognitive processes during AI-assisted language learning activities. Furthermore, investigating the synergistic effects of combining different AI-assisted language learning strategies, such as personalized feedback with virtual reality simulations or natural language processing technologies, can provide a more holistic approach to language learning.

Conclusion

In summary, this study has illuminated the potential of AI-assisted language learning strategies in mitigating extraneous cognitive load and optimizing language learning experiences. The findings carry significant theoretical implications, shedding light on the impact of these strategies within the context of language learning. Furthermore, they hold pedagogical significance for language instructors and curriculum designers aiming to elevate instructional practices through the integration of AI technologies. However, this research is not without its limitations. The study's sample size was relatively small and limited to a specific demographic, which may affect the generalizability of the results. Additionally, the study relied on self-reported data, which can be subject to biases. The duration of the intervention was also relatively short, which may not fully capture the long-term effects of AI-assisted language learning strategies. Despite these limitations, this study paves the way for future research endeavors to investigate the effects of AI-assisted language learning strategies in diverse settings and among varied learner populations. Such future research can contribute to the progression of technology-enhanced language education by addressing these limitations and exploring new dimensions of AI integration in language learning.

Appendix A

Cognitive Load Scale Items and Descriptions by Group.

No	Group	Item Description	M Pre	M Post	SD Pre	SD Post	Extracted Themes	Pedagogical Insights
1	Group I	As a result of employing AI- assisted personalized feedback and adaptive learning strategies during the course, I invested cognitive load in receiving tailored feedback and adapting my learning approach	4.25	6.54	1.784	1.652	Personalized feedback, adaptive learning	Personalized feedback and adaptive strategies significantly reduce cognitive load, enabling more effective learning adjustments
2	Group I	During the course, I found myself investing cognitive load in utilizing AI- assisted personalized feedback and adaptive learning strategies to enhance my learning outcomes	4.25	6.54	1.784	1.652	Active utilization, cognitive investment	Active engagement with AI-assisted feedback improves learning outcomes by efficiently managing cognitive resources

(continued)

(continued)

No	Group	Item Description	M Pre	M Post	SD Pre	SD Post	Extracted Themes	Pedagogical Insights
3	Group 1	The AI-assisted personalized feedback and adaptive learning strategies employed during the course required me to invest _____ cognitive load in adjusting my learning strategies based on personalized feedback	4.25	6.54	1.784	1.652	Strategy adjustment, feedback integration	Adapting learning strategies based on AI feedback optimizes cognitive load and enhances learning efficiency
4	Group 2	As a result of employing AI-assisted interactive exercises during the course, I invested _____ cognitive load in actively engaging with interactive exercises to enhance my understanding	4.25	6.54	1.784	1.652	Interactivee engagement, understanding enhancement	Interactive exercises facilitated by AI significantly enhance cognitive engagement and understanding of course material
5	Group 2	During the course, I found myself investing _____ cognitive load in utilizing AI-assisted interactive exercises to actively participate and reinforce my learning	4.15	6.54	1.789	1.652	Active participation, learning reinforcement	Active participation in AI-assisted exercises reinforces learning and promotes deeper engagement with content

(continued)

(continued)

No	Group	Item Description	M Pre	M Post	SD Pre	SD Post	Extracted Themes	Pedagogical Insights
6	Group 2	The AI-assisted interactive exercises employed during the course required me to invest _____ cognitive load in actively solving problems and engaging with interactive learning materials	5.25	6.66	1.884	1.852	Problem solving, interactive learning	Solving problems through AI-assisted exercises enhances cognitive load management and learning outcomes
7	Group 3	As a result of employing AI-assisted speech recognition strategies during the course, I invested _____ cognitive load in practicing and improving my spoken language skills through speech recognition technology	5.25	6.66	1.884	1.852	Speech recognition, skill improvement	Speech recognition technology aids in practicing and improving spoken language skills by managing cognitive load effectively
8	Group 3	During the course, I found myself investing _____ cognitive load in utilizing AI-assisted speech recognition strategies to practice and improve my spoken language skills through speech recognition technology	5.25	6.66	1.884	1.852	Pronunciation practice, oral communication	Utilizing speech recognition for pronunciation practice enhances oral communication skills and reduces cognitive load

(continued)

(continued)

No	Group	Item Description	M Pre	M Post	SD Pre	SD Post	Extracted Themes	Pedagogical Insights
9	Group 3	The AI-assisted speech recognition strategies employed during the course required me to invest _____ cognitive load in practicing and improving my spoken language skills through speech recognition technology	5.25	6.66	1.884	1.852	Speaking practice, feedback reception	AI-assisted speech recognition for speaking practice and feedback significantly improves spoken language proficiency

Appendix B

Evaluation of AI-Assisted Intervention for Cognitive Load and Language Learning Outcomes.

Question	Mean	SD	Sample Student Responses	Extracted Themes+Pedagogical Insights
1. Did you find the AI-assisted intervention helpful in improving your language learning outcomes?	4.95	0.228	"The AI tools really boosted my vocabulary and grammar understanding." (S1) "I noticed a significant improvement in my writing skills." (S2) "my speaking confidence increased thanks to the AI feedback." (S3)	Enhanced language proficiency; effective tool for vocabulary, grammar, and writing improvement; increased speaking confidence
2. Did you find the AI-assisted intervention helpful in managing cognitive load during the study?	4.67	0.234	"It was easier to focus on learning without feeling overwhelmed." (S2) "the AI helped me break down complex topics into manageable parts." (S4) "I felt less stressed and more organized." (S5)	Reduced cognitive load; improved focus and concentration; better organization and stress management

(continued)

(continued)

Question	Mean	SD	Sample Student Responses	Extracted Themes+Pedagogical Insights
3. Did you find the AI-assisted intervention causing cognitive offloading?	4.89	0.220	"The AI took care of the repetitive tasks, so I could focus on understanding concepts." (S3) "I didn't have to worry about minor errors, which saved me time." (S6) "the automation of routine tasks was a huge relief." (S7)	Cognitive offloading; AI handling repetitive tasks allows deeper conceptual understanding; time-saving and reduced worry about minor errors
4. Did you feel that the AI-assisted intervention was personalized to your learning needs and preferences?	4.95	0.228	"The lessons felt tailored to my pace and interests." (S4) "I appreciated how the AI adapted to my progress." (S8) "the personalized feedback was very motivating." (S9)	Personalized learning; customization based on individual pace and interests; adaptive learning and motivating feedback
5. Did you find the AI-assisted intervention easy to navigate and use for learning purposes?	4.95	0.228	"The interface was user-friendly and intuitive." (S5) "I had no trouble finding the resources I needed." (S10) "the design made it easy to stay on track with my studies." (S11)	User-friendly interface; ease of navigation enhances learning experience; design supports effective study habits
6. Did you feel that the AI-assisted intervention helped you manage cognitive load in your language learning journey?	4.67	0.234	"I didn't feel as mentally exhausted after using the AI tools." (S6) "the AI helped me prioritize my tasks better." (S12) "I could balance my studies with other activities more easily." (S13)	Reduced cognitive load; better task prioritization; enhanced balance between studies and other activities
7. Did you find the AI-assisted intervention engaging and motivating?	4.80	0.210	"The interactive elements kept me engaged." (S7) "I felt more motivated to learn with the AI's encouragement." (S14) "the gamified aspects made studying fun." (S15)	Increased engagement; motivational support; gamification enhances learning experience
8. Would you recommend the AI-assisted intervention to other students?	4.90	0.215	"I would definitely recommend it to my peers." (S8) "It's a great tool for anyone looking to improve their language skills." (S16) "I think everyone could benefit from using this AI." (S17)	High recommendation; perceived as beneficial for language improvement; positive peer endorsement

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Data Availability Statement

The data will be made available on reasonable request.

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